

Shreya Nair

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SUMMARY

Data Scientist with 2 years of experience building and deploying machine learning solutions and scalable data pipelines. Developed an embedding-driven recommendation engine that boosted conversion rates and engineered an NLP tagging system for over 200,000 products with zero errors. Proficient in Python, SQL, Docker, Kubernetes and CI/CD on cloud platforms, seeking to drive actionable insights for customer-centric analytics.

EXPERIENCE

Think Dirty - Clean Beauty App

Data Science Specialist

Toronto, Canada · Jan 2025 - Present

- Developed and deployed an embedding-driven recommendation engine using PyTorch and SageMaker, increasing conversion rates and demonstrating predictive modeling that supports customer goals
- Streamlined SQL and Spark pipelines, building robust analytical datasets and dashboards that accelerated decision-making across compliance workflows
- Delivered end-to-end ML solutions with containerized deployments (Docker, Kubernetes) and CI/CD practices, reducing deployment cycles from 2 weeks to 3 days and achieving 9% production uptime

Data Science Intern

Toronto, Canada · Aug 2024 - Dec 2024

- Engineered an NLP-powered tagging system for 200k+ products using transformers, achieving zero error margin and improving algorithmic accuracy and efficiency
- Deployed real-time recommendation and anomaly detection models on Azure ML and Databricks (Spark), streamlining model delivery and increasing prediction reliability
- Analyzed application event logs with BigQuery and SQL, generating automated analytics reports that drove a 15% improvement in feature adoption and identified performance bottlenecks

Claysys Technologies

Project Coordinator – RPA

Kochi, India · Apr 2021 - Nov 2022

- Utilized SQL and pandas to extract and analyze operational data, developing process-automation solutions that reduced manual reporting effort for credit union clients
- Designed Power BI dashboards to visualize key performance indicators, enhancing stakeholder insights and decision-making
- Collaborated with cross-functional teams using Agile and Jira to map business processes, identify inefficiencies, and recommend data-driven improvements that streamlined operations

EDUCATION

PGP, Artificial Intelligence and Machine Learning

May 2023 - Dec 2024

Lambton College

Dual Degree Masters, Computer Science, Data Science and AI

Aug 2015 - Dec 2020

SCMS School of technology and Management

PROJECTS

Realty Robo – AI-Powered Real Estate Chatbot

- Developed an intelligent Python-based chatbot leveraging LLaMA-2 and prompt engineering for natural language understanding, enabling users to inquire about property listings, schedule appointments, and receive financial advice.
- Implemented a three-pronged approach: intent classification using LSTM, entity extraction, and response generation, enhancing user interaction efficiency.
- Optimized the chatbot's performance through fine-tuning on custom datasets, resulting in a 30% improvement in response accuracy compared to traditional methods.
- Deployed the solution on Google Cloud Platform, ensuring scalability and reliability for real-time user engagement.
- Containerized the chatbot using Docker and implemented CI/CD pipelines with GitHub Actions to ensure seamless deployments and rapid iteration.

London Borough of Barnet Anomaly Detection

- Utilized Python and SQL to ingest, clean, and analyze corporate credit card transactions, uncovering spending patterns and irregularities across multiple departments.
- Applied Isolation Forest algorithm on financial transactions, uncovering anomalies indicative of potential fraud or irregularities.
- Developed interactive dashboards with Plotly Dash and Power BI to visualize transaction trends and highlight anomalies, facilitating data-driven oversight for finance stakeholders.
- Deployed the dashboards on Azure App Service to provide secure, scalable access for cross-functional teams and support ongoing monitoring.

Stock Market Sentiment Analysis

- Developed a predictive model integrating sentiment analysis of financial news and social media with time series data to forecast stock closing prices.
- Implemented Natural Language Processing techniques to extract sentiment features from textual data, enhancing the model's understanding of market moods.
- Utilized Generative Adversarial Networks (GANs) to generate realistic stock price sequences, improving prediction accuracy over traditional models.
- Conducted comprehensive data preprocessing, including feature extraction and normalization, to prepare datasets for model training and evaluation.

SKILLS

- **Programming Languages:** Python, SQL
- **Libraries & Frameworks:** Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, Statsmodels, Matplotlib, Seaborn, Plotly-, Transformers (Hugging Face)
- **Cloud Platforms:** Microsoft Azure (Azure ML, Azure Data Factory, Azure Databricks), Google Cloud Platform (BigQuery), AWS (S3, SageMaker)
- **Data Visualization Reporting:** Power BI, Tableau, Plotly Dash, Gradio, Seaborn
- **Machine Learning & Statistical Techniques:** Supervised Learning, Unsupervised Learning, Time Series Analysis, Predictive Modeling, Statistical Analysis, Model Evaluation & Optimization, Feature Engineering, Natural Language Processing (NLP), Large Language Models (LLMs), Prompt Engineering
- **Tools & Platforms:** Jupyter Notebook, Visual Studio Code, Git, GitHub, Anaconda, RStudio, SQL Server Management Studio, pgAdmin, TestLodge, Streamlit, REST APIs, JSON, Jira
- **DevOps & MLOps:** Docker, Git, CI/CD, Kubernetes
- **Big Data Tools:** Hadoop, Spark
- **Soft Skills:** Effective Communication, Cross-Functional Collaboration, Problem-Solving, Strategic Thinking, Continuous Learning, Mentorship, Adaptability

CERTIFICATIONS

- Introduction to AI and ML on Google Cloud: https://www.cloudskillsboost.google/public_profiles/bd74940e-bce3-41f7-9726-af310d3bac4e/badges/9081796?utm_medium=social&utm_source=twitter
- AI - Data Scientist: [https://s3.ap-south-1.amazonaws.com/insdms2.0/SSC-Certificates/2025/TPA-\[L5-7\]-Regular/TPA0810420250819/abc4130.pdf](https://s3.ap-south-1.amazonaws.com/insdms2.0/SSC-Certificates/2025/TPA-[L5-7]-Regular/TPA0810420250819/abc4130.pdf)
- Deep Learning: Tensorflow and Keras: <https://courses.cognitiveclass.ai/certificates/0ecf58bc2ea445e4b26e44b8db76ae9b>
- Generative AI: The Complete Generative AI on Google Cloud: https://www.cloudskillsboost.google/public_profiles/bd74940e-bce3-41f7-9726-af310d3bac4e

EXTRA CONTRIBUTIONS

Stem Fellowship

Outreach Team Member: Data Science Committee

- Mentorship in Data Science & AI/ML Projects
- Workshop Facilitation
- Big Data and AI Challenge (BDC) Support
- Technical Writing & Presentation Guidance
- Project Review & Feedback
- Youth Engagement and Outreach